02 Gravitation

The Discovery of the Law of Gravitation

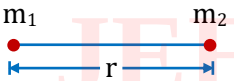
The way the law of universal gravitation was discovered is often considered the paradigm of modern scientific technique. The major steps involved were

- The hypothesis about planetary motion given by **Nicolaus Copernicus (1473–1543)**.
- The careful experimental measurements of the positions of the planets and the Sun by **Tycho Brahe (1546–1601)**.
- Analysis of the data and the formulation of empirical laws by **Johannes Kepler (1571–1630)**.
- The development of a general theory by **Isaac Newton (1642–1727)**.

Universal Law of Gravitation: Newton's Law

According to this law "Each particle attracts every other particle. The force of attraction between them is directly proportional to the product of their masses and inversely proportional to square of the distance between them".

$$F \propto \frac{m_1 m_2}{r^2} \quad \text{or} \quad F = G\frac{m_1 m_2}{r^2}$$



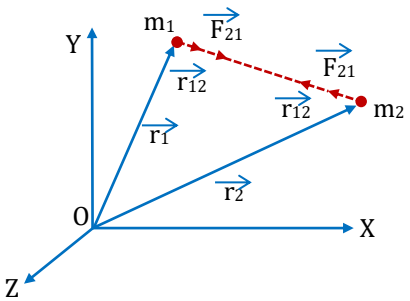
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where $G = 6.67 \times 10^{-11}\ Nm^2\ kg^{-2}$ is the universal gravitational constant.

Dimensional Formula of G :

$$G = \frac{Fr^2}{m_1 m_2} = \frac{[MLT^{-2}][L^2]}{[M^2]} = [M^{-1}L^3T^{-2}]$$

Vector Form of Newton's Law of Gravitation



Let $\vec{r}_{12}$ = Displacement vector from m_1 to m_2

$\vec{r}_{21}$ = Displacement vector from m_2 to m_1

$\vec{F}_{21}$ = Gravitational force exerted on m_2 by m_1

$\vec{F}_{12}$ = Gravitational force exerted on m_1 by m_2

$$\vec{F}_{12} = -\frac{Gm_1m_2}{r_{21}^2}\hat{r}_{21} = -\frac{Gm_1m_2}{r_{21}^3}\vec{r}_{21}$$

Negative sign shows that :

(i) The direction of $\vec{F}_{12}$ is opposite to that $\vec{r}_{21}$

(ii) The gravitational force is attractive in nature

Similarly, $\vec{F}_{21} = -\frac{Gm_1m_2}{r_{12}^2}\hat{r}_{12}$ or $\vec{F}_{21} = -\frac{Gm_1m_2}{r_{12}^3}\vec{r}_{12} \Rightarrow \vec{F}_{12} = -\vec{F}_{21}$

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The gravitational force between two bodies are equal in magnitude and opposite in direction.

Important characteristics of gravitational force:

(i) Gravitational force between two bodies form an action and reaction pair i.e. the forces are equal in magnitude but opposite in direction.

(ii) Gravitational force is a central force i.e. it acts along the line joining the centers of the two interacting bodies.

(iii) Gravitational force between two bodies is independent of the nature of the medium, in which they lie.

(iv) Gravitational force between two bodies does not depend upon the presence of other bodies.

(v) Gravitational force is negligible in case of light bodies but becomes appreciable in case of massive bodies like stars and planets.

(vi) Gravitational force is long range-force i.e., gravitational force between two bodies is effective even if their separation is very large.

For Illustration gravitational force between the Sun and the Earth is of the order of 10^{27} *N* although distance between them is 1.5×10^{7} *km.*

Illustration 1:

Two particles of masses 1 *kg* and 2 *kg* are placed at a separation of 50 *cm*. Assuming that the only forces acting on the particles are their mutual gravitation, find the initial acceleration of heavier particle.

Solution:

Force exerted by one particle on another, $F = \frac{Gm_1m_2}{r^2} = \frac{6.67\times10^{-11}\times1\times2}{(0.5)^2} = 5.3\times10^{-10}\,N$

Acceleration of heavier particle = $\frac{F}{m_2} = \frac{5.3\times10^{-10}}{2} = 2.65\times10^{-10}\,ms^{-2}$

This Illustration shows that gravitation is very weak but only this force keep bind our solar system and also this universe, all galaxies and other interstellar system.

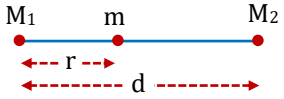
Illustration 2:

Two stationary particles of masses M_1 and M_2 are at a distance *'d'* apart. A third particle lying on the line joining the particles, experiences no resultant gravitational forces. What is the distance of this particle from M_1.

Solution:

The force on *m* towards M_1 is $F_1 = \frac{GM_1m}{r^2}$

The force on *m* towards M_2 is $F_2 = \frac{GM_2m}{(d-r)^2}$



According to question, net force on *m* is zero i.e. $F_1 = F_2$

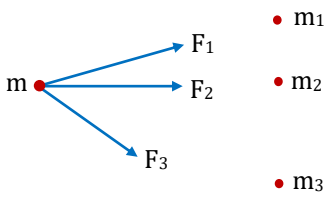
$$\Rightarrow \quad \frac{GM_1m}{r^2} = \frac{GM_2m}{(d-r)^2} \qquad \Rightarrow \quad \left(\frac{d-r}{r}\right)^2 = \frac{M_2}{M_1}$$

$$\Rightarrow \quad \frac{d}{r} - 1 = \frac{\sqrt{M_2}}{\sqrt{M_1}} \qquad \Rightarrow \quad r = d\left[\frac{\sqrt{M_1}}{\sqrt{M_1}+\sqrt{M_2}}\right]$$

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Principle of superposition

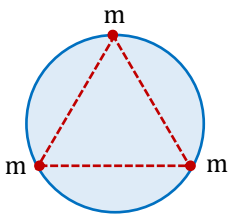
The force exerted by a particle on other particle remains unaffected by the presence of other nearby particles in space. Total force acting on a particle is the vector sum of all the forces acted upon by the individual masses when they are taken alone.



$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + ...$$

Illustration 3:

Three particles, each of mass *m*, are situated at the vertices of an equilateral triangle of side *'a'*. The only forces acting on the particles are their mutual gravitational forces. It is desired that each particle moves in a circle while maintaining their original separation *'a'*. Determine the initial velocity that should be given to each particle and time period of circular motion.



Solution:

The resultant force on particle at *A* due to other two particles is

$$F_A = \sqrt{F_{AB}^2 + F_{AC}^2 + 2F_{AB}F_{AC}\cos 60°} = \sqrt{3}\frac{Gm^2}{a^2} \quad ...(i) \quad \left[\because F_{AB} = F_{AC} = \frac{Gm^2}{a^2}\right]$$

Radius of the circle $r = \frac{2}{3} \times a \sin 60° = \frac{a}{\sqrt{3}}$

If each particle is given a tangential velocity *v*,
so that *F* acts as the centripetal force,

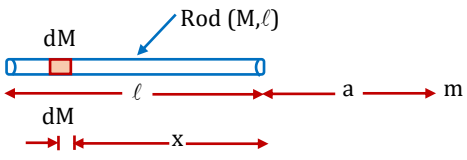
Now $\frac{mv^2}{r} = \sqrt{3}\frac{mv^2}{a}$ ...(ii)

From (i) and (ii) $\sqrt{3}\frac{mv^2}{a} = \frac{Gm^2\sqrt{3}}{a^2} \Rightarrow v = \sqrt{\frac{Gm}{a}}$

Time period, $T = \frac{2\pi r}{v} = \frac{2\pi a}{\sqrt{3}}\sqrt{\frac{a}{Gm}} = 2\pi\sqrt{\frac{a^3}{3Gm}}$

Illustration 4:

Find gravitational force exerted by point mass *'m'* on uniform rod (mass *'M'* and length 'ℓ').



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Solution:

dF = force on element in horizontal direction = $\frac{G.dM.m}{(x+a)^2}$

where $dM = \frac{M}{\ell}dx$.

$$\therefore \quad F = \int dF = \int_0^{\ell} \frac{G.Mmdx}{\ell(x+a)^2} = \frac{G.Mm}{\ell}\int_0^{\ell} \frac{dx}{(x+a)^2} = \frac{G.Mm}{\ell}\left[-\frac{1}{(\ell+a)} + \frac{1}{a}\right] = \frac{GMm}{(\ell+a)a}$$

Gravitational Field

The space surrounding the body within which its gravitational force of attraction is experienced by other bodies is called gravitational field. Gravitational field is very similar to electric field in electrostatics where charge *'q'* is replaced by mass *'m'* and electric constant *'K'* is replaced by gravitational constant *'G'*. The intensity of gravitational field at a point is defined as the force experienced by a unit mass placed at that point.

$$\vec{E} = \frac{\vec{F}}{m}$$

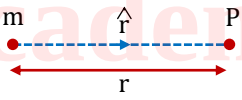
The unit of the intensity of gravitational field is $N\,kg^{-1}$.

Intensity of Gravitational Field due to Point Mass

The force due to mass m on test mass m_0 placed at point P is given by :

$$F = \frac{GMm_0}{r^2}$$

Hence $E = \frac{F}{m_0} \Rightarrow E = \frac{GM}{r^2}$



In vector form, $\vec{E} = -\frac{GM}{r^2}\hat{r}$

Dimensional formula of intensity of gravitational field = $\frac{F}{m} = \frac{[MLT^{-2}]}{[M]} = [M^0LT^{-2}]$

Illustration 5:

Find the distance of a point from the Earth's centre where the resultant gravitational field due to the Earth and the moon is zero. The mass of the Earth is 6.0×10^{24} *kg* and that of the moon is 7.4×10^{22} *kg*. The distance between the Earth and the moon is 4.0×10^5 *km*.

Solution:

The point must be on the line joining the centres of the Earth and the moon and in between them. If the distance of the point from the Earth is x, the distance from the moon is $(4.0 \times 10^5\ km - x)$. The magnitude of the gravitational field due to the Earth is

$$E_1 = \frac{GM_e}{x^2} = \frac{G \times 6 \times 10^{24}\,kg}{x^2}$$

and magnitude of the gravitational field due to the moon is

$$E_2 = \frac{GM_m}{(4.0 \times 10^5\,km - x)^2} = \frac{G \times 7.4 \times 10^{22}\,kg}{(4.0 \times 10^5\,km - x)^2}$$

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These fields are in opposite directions. For the resultant field to be zero $E_1 = E_2$.

or, $$\frac{6\times10^{24}\,kg}{x^2}=\frac{7.4\times10^{22}\,kg}{(4.0\times10^5\,km-x)^2}$$

or, $$\frac{x}{4.0\times10^5\,km-x}=\sqrt{\frac{6\times10^{24}}{7.4\times10^{22}}}=9$$

or, $x = 3.6 \times 10^5\ km.$

Illustration 6:

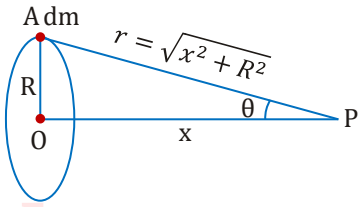
Calculate gravitational field intensity due to a uniform ring of mass M and radius R at a distance x on the axis from center of ring.

Solution:

Consider any particle of mass dm. Gravitational field at point P due to dm

$$dE=\frac{Gdm}{r^2}\text{ along }PA$$

Component along PO is $dE\cos\theta = \frac{Gdm}{r^2}\cos\theta$



Net gravitational field at point P is

$$E=\int\frac{Gdm}{r^2}\cos\theta=\frac{G\cos\theta}{r^2}\int dm$$

$$=\frac{GMx}{(R^2+x^2)^{3/2}}\text{ towards the center of ring}$$

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Illustration 7:

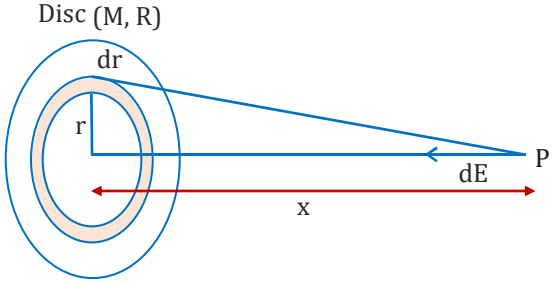
Calculate gravitational field intensity at a distance x on the axis from centre of a uniform disc of mass M and radius R.

Solution:

Consider a elemental ring of radius r and thickness dr on surface of disc as shown in figure.

Gravitational field due to elemental ring

$$dE=\frac{G(dM)x}{(x^2+r^2)^{3/2}}$$



Here $$dM=\frac{M}{\pi R^2}.2\pi rdr=\frac{2M}{R^2}rdr$$

$\therefore$ $$dE=\frac{G.2Mxrdr}{R^2(x^2+r^2)^{3/2}}$$

$\therefore$ $$E=\int_0^R\left(\frac{2GMx}{R^2}\right)\frac{rdr}{(x^2+r^2)^{3/2}}$$

$\therefore$ $$E=\frac{2GMx}{R^2}\left[\frac{1}{x}-\frac{1}{\sqrt{x^2+R^2}}\right]$$

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Illustration 8:

For a given uniform spherical shell of mass M and radius R, find gravitational field at a distance r from centre in following two cases:

(a) $r \geq R$ (b) $r < R$

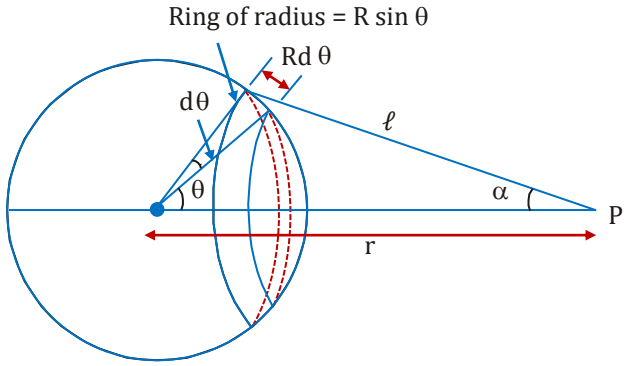
Solution:

$$dE = \frac{GdM}{\ell^2}.\cos\alpha \qquad r \geq R$$

$$dM = \frac{M}{4\pi R^2} \times 2\pi R \sin\theta R d\theta$$

$$dM = \frac{M}{2}\sin\theta\, d\theta$$

$$\therefore \quad dE = \frac{GM\sin\theta\cos\alpha\, d\theta}{2\ell^2}$$



Now $\ell^2 = R^2 + r^2 - 2rR\cos\theta$ ...(i)

$R^2 = \ell^2 + r^2 - 2\ell r\cos\alpha$ ...(ii)

$$\therefore \quad \cos\alpha = \frac{\ell^2 + r^2 - R^2}{2\ell r}$$

$$\cos\theta = \frac{R^2 + r^2 - \ell^2}{2rR}$$

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Differentiating (1)

$$\therefore \quad 2\ell\, d\ell = 2rR\sin\theta\, d\theta$$

$$\therefore \quad dE = \frac{GM}{2\ell^2}\cdot\frac{\ell\, d\ell}{Rr}\cdot\frac{\ell^2 + r^2 - R^2}{(2\ell r)} \qquad \Rightarrow \quad dE = \frac{GM}{4Rr^2}\left[1 + \frac{r^2 - R^2}{\ell^2}\right]d\ell$$

$$\therefore \quad E = \int dE = \frac{GM}{4Rr^2}\left[\int_{r+R}^{r-R} d\ell + (r^2 - R^2)\int_{r+R}^{r-R}\frac{d\ell}{\ell^2}\right] \qquad \Rightarrow E = \frac{GM}{r^2}, r \geq R$$

If point is inside the shell limit changes to $[(R - r)\ to\ R + r]$

$E = 0$ when $r < R$.

Illustration 9:

Find the relation between the gravitational field on the surface of two planets A and B of masses m_A, m_B and radius R_A and R_B respectively if

(i) they have equal mass (ii) they have equal (uniform) density

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Solution:

Let E_A and E_B be the gravitational field intensities on the surface of planets A and B.

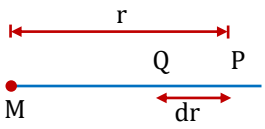
then, $E_A = \frac{Gm_A}{R_A^2} = \frac{G\frac{4}{3}\pi R_A^3 \rho_A}{R_A^2} = \frac{4G\pi}{3}\rho_A R_A$

Similarly, $E_B = \frac{GM_B}{R_{B^2}} = \frac{4G}{3}\pi\ \rho_B R_B$

(i) For $m_A = m_B, \frac{E_A}{E_B} = \frac{R_B^2}{R_A^2}$ (ii) For $\rho_A = \rho_B, \frac{E_A}{E_B} = \frac{R_A}{R_B}$

Gravitational Potential

The gravitational potential at a point in the gravitational field of a body is defined as the amount of work done by an external agent in bringing a body of unit mass from infinity to that point, slowly (no change in kinetic energy). Gravitational potential is very similar to electric potential in electrostatics.



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Gravitational Potential Due to a Point Mass

Let the unit mass be displaced through a distance dr towards mass M, then work done is given by

$$dW = F\,dr = \frac{GM}{r^2}dr$$

Total work done in displacing the particle from infinity to point P is

$$W = \int dw = \int_{\infty}^{r} \frac{GM}{r^2}dr = \frac{-GM}{r}$$

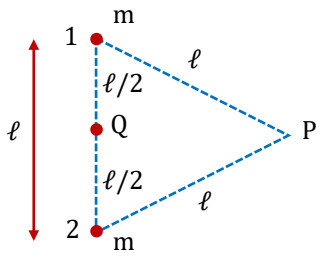
Thus, gravitational potential, $\boxed{V = -\frac{GM}{r}}$.

The unit of gravitational potential is $J\ kg^{-1}$. Dimensional Formula of gravitational potential

$$= \frac{\text{Work}}{\text{Mass}} = \frac{[ML^2T^{-2}]}{[M]} = [M^0L^2T^{-2}]$$

Illustration 10:

Find out potential at P and Q due to the two point mass system. Find out work done by external agent in bringing unit mass from P to Q. Also find work done by gravitational force.



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Solution:

(i) V_{P1} = potential at P due to mass 'm' at '1' = $-\frac{Gm}{\ell}$

$V_{P2} = -\frac{Gm}{\ell}$

$\therefore \quad V_P = V_{P1} + V_{P2} = -\frac{2Gm}{\ell}$

(ii) $V_{Q1} = -\frac{Gm}{\ell/2} \quad \Rightarrow \quad V_{Q2} = -\frac{Gm}{\ell/2}$

$\therefore \quad V_Q = V_{Q1} + V_{Q2} = -\frac{Gm}{\ell/2} - \frac{Gm}{\ell/2} = -\frac{4Gm}{\ell}$

Force at point $Q = 0$

(iii) work done by external agent = $(V_Q - V_P) \times 1 = -\frac{2GM}{\ell}$

(iv) work done by gravitational force = $V_P - V_Q = \frac{2GM}{\ell}$

Illustration 11:

Find potential at a point 'P' at a distance 'x' on the axis away from centre of a uniform ring of mass M and radius R.

Solution:

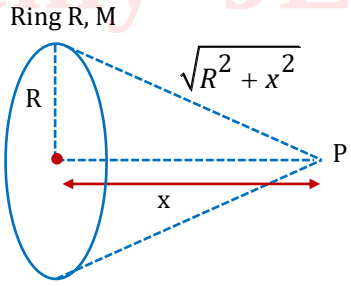
Ring can be considered to be made of large number of point masses (m_1, m_2 ..... etc)

$$V_P = -\frac{Gm_1}{\sqrt{R^2+x^2}} - \frac{Gm_2}{\sqrt{R^2+x^2}} - .....$$

$$= -\frac{G}{\sqrt{R^2+x^2}}(m_1 + m_2 .....) = \frac{GM}{\sqrt{R^2+x^2}},$$

where $M = m_1 + m_2 + m_3 + ...$

Potential at centre of ring = $-\frac{G.M}{R}$



Relation between Gravitational Field and Potential

The work done by an external agent to move unit mass from a point to another point in the direction of the field E, slowly through an infinitesimal distance dr = Force by external agent × Distance moved = $-Edr$.

Thus $dV = -Edr$

$\Rightarrow \quad E = -\frac{dV}{dr}$

Therefore, gravitational field at any point is equal to the negative gradient at that point.

Illustration 12:

The gravitational field in a region is given by $\vec{E} = 20(\hat{i} + \hat{j})N/kg$. Find the gravitational potential at the origin (0, 0) ($in\ J/kg$)

(A) Zero (B) $20\sqrt{2}$ (C) $-20\sqrt{2}$ (D) can not be defined

Ans. (A)

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Solution:

$$V = -\int E.dr = \left[\int E_x.dx + \int E_y.dy\right] = 20x + 20y$$

At origin $V = 0$.

Illustration 13:

In above problem, find the gravitational potential at a point whose co-ordinates are (5, 4) : (*in J/kg*)

(A) – 180 (B) 180 (C) – 90 (D) zero

Ans. (B)

Solution:

$V = 20 \times 5 + 20 \times 4 = 180\ J/kg$

Illustration 14:

In the above problem, find the work done in shifting a particle of mass 1 *kg* from origin (0, 0) to a point (5, 4)*:* (In *J*)

(A) – 180 (B) 180 (C) – 90 (D) zero

Ans. (B)

Solution:

$W = m\ (V_f - V_i) = 1\ (180 - 0) = 180\ J$

Illustration 15:

Find gravitational field at a point (x, y, z).

$v = 2x^2 + 3y^2 + zx$,

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Solution:

$$\vec{E} = -\nabla V$$

$$E_x = \frac{-\partial v}{\partial x} = -4x - z \quad ; \quad E_y = -6y \quad ; \quad E_z = -x$$

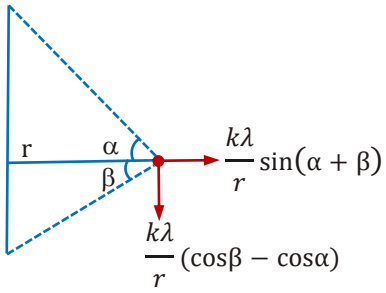
$\therefore$ Field $= \vec{E} = -[(4x + z)\,\hat{\imath} + 6y\hat{\jmath} + x\hat{k}]$

Analogy between Electrostatics and Gravitation

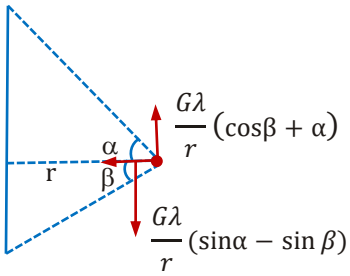
(1)	**Point Charge**	**Point Mass**
	(a) $E = \frac{kQ}{r^2}$	$g = \frac{GM}{r^2}$
	(b) $V = \frac{kQ}{r}$	$V_G = \frac{-GM}{r}$
(2)	**Uniform charged ring**	**Ring of uniform mass distribution**
	(a) $E = \frac{kQx}{(r^2 + x^2)^{3/2}}$ on axis	$g = \frac{GMx}{(r^2 + x^2)^{3/2}}$ on axis
	E is max. when $x = \frac{r}{\sqrt{2}}$	g is max. when $x = \frac{r}{\sqrt{2}}$
	(b) $V = \frac{kQ}{\sqrt{r^2 + x^2}}$ on axis, $\frac{kQ}{r}$ at center	$V_G = \frac{-GM}{\sqrt{r^2 + x^2}}$ on axis, $\frac{-GM}{r}$ at center

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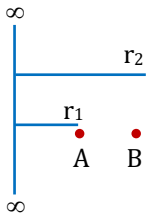
(3) **Uniform linear charge**



Uniform linear mass



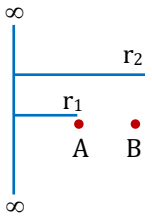
(4) **Infinite Linear charge**



(a) $E=\dfrac{2K\lambda}{r}$

(b) $V_B-V_A=-2K\lambda\ln\dfrac{r_2}{r_1}$

Infinite linear mass



$g=\dfrac{2G\lambda}{r}$

$V_B-V_A=2G\lambda\ln\left(\dfrac{r_2}{r_1}\right)$

(5) **Infinite Sheet of charge**

$E=\dfrac{\sigma}{2\varepsilon_0}$

$=2\pi K\sigma$

Infinite Sheet of mass

$g=\dfrac{\sigma}{2}\times4\pi G=2\pi G\sigma$

(σ = Mass per unit area)

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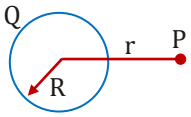
- Notice gravitational force is always attractive and hence gravitational potential is always *–ve* (for a repulsive force potential is positive). This can be explained from the sign of W_{ext} in moving the test charge from ∞ to the point under consideration.
- Since $\vec{g}$ points from *B* towards *A* potential increases as we move from *A* to *B*. Just like electric potential, gravitational potential also increases opposite to field direction.

(6) **Uniformly charged hollow sphere**

Charge *Q*, radius *R*

distance of field point from center *r*

Case I. $r > R$



$E=\dfrac{kQ}{r^2}$

$V=\dfrac{kQ}{r}$

Case II. $r < R$



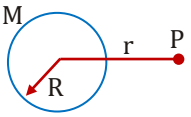
$E=0$

$V=\dfrac{kQ}{R}$

Hollow sphere of uniform mass

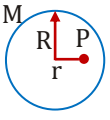
Mass *M*, radius *R*

distance of field point from center *r*



$g=\dfrac{GM}{r^2}$

$V_G=-\dfrac{GM}{r}$



$g=0$

$V_G=-\dfrac{GM}{R}$

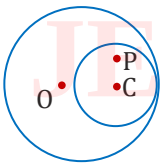
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(7)	**Electrostatics self energy of uniformly charged thin spherical shell.**	**Gravitational self energy of uniform thin spherical shell.**
	$U=\frac{KQ^2}{2R}$	$U=-\frac{GM^2}{2R}$
(8)	**Uniformly charged solid sphere** mass M, radius R	**Uniformly solid sphere** mass M, radius R
	$E=\frac{kQ}{r^2}, r>R$	$g=\frac{GM}{r^2}, r>R$
	$\frac{kQr}{R^3}, r<R$	$\frac{GM}{R^3}r, r<R$
	$V=\frac{kQ}{r}, r>R$	$V_a=-\frac{GM}{r}, r>R$
	$\frac{kQ}{2R^3}\left(3R^2-r^2\right), r<R$	$\frac{-GM}{2R^3}\left(3R^2-r^2\right), r<R$
(9)	**Electrostatics self energy of uniformly charged solid sphere.**	**Gravitational self energy of uniform solid sphere.**
	$U=\frac{3}{5}\frac{KQ^2}{R}$	$U=-\frac{3}{5}\frac{GM^2}{R}$

Illustration 16:

A uniform solid sphere of density ρ and radius R has a spherical cavity of radius r inside it as shown. Find gravitation field at

(a) O (b) C (c) P (prove that field inside cavity is uniform)



Ans. (a) $\frac{k\left(\frac{4}{3}\pi r^3\rho\right)}{(OC)^2}\hat{CO}$ (b) $\frac{4\pi G\rho\overrightarrow{OC}}{3}$ (c) $\frac{4}{3}\pi G\rho\overrightarrow{OC}$

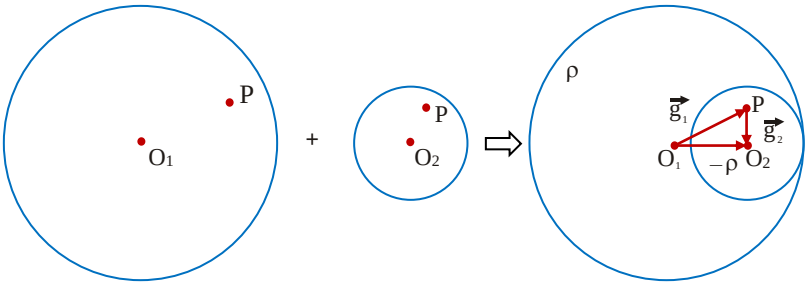
Solution:

$\vec{g}$ = Gravitational field at any point inside sphere

$$\vec{g}=\frac{GM}{R^3}\vec{r}=\frac{G}{R^3}\frac{4}{3}\pi R^3\rho\vec{r}$$

$$\vec{g}=\frac{4}{3}\pi G\rho\vec{r}$$

Let the sphere with cavity is formed by superimposing it with a small sphere of density (−ρ) as shown



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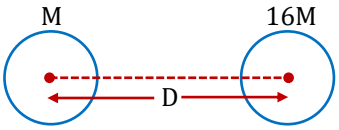
Resultant field $\vec{g}=\vec{g}_1+\vec{g}_2$

$$=\left(\frac{4}{3}\pi G\rho\right)\overrightarrow{O_1P}+\left(\frac{4}{3}\pi G\rho\right)\overrightarrow{PO_2}$$

$$=\frac{4}{3}\pi G\rho\overrightarrow{O_1O_2} \quad \left[\overrightarrow{O_1O_2}=\overrightarrow{OC}\right]$$

It is independent of position of point inside cavity.

Illustration 17:



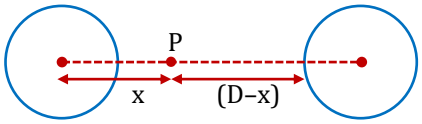
(a) Find where is gravitational field intensity is zero.

(b) Find gravitational potential at *P*.

Solution:

(a) $\frac{GM}{x^2}=\frac{G(16M)}{(D-x)^2}$

$4=\frac{D-x}{x} \Rightarrow x=\frac{D}{5}$

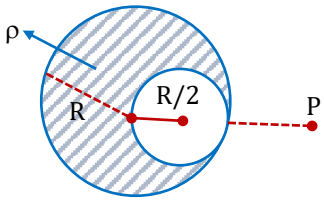


(b) $v_P=\frac{-GM}{\frac{D}{5}}+\left(\frac{-G(16M)}{\frac{4D}{5}}\right)=-\frac{25GM}{D}$

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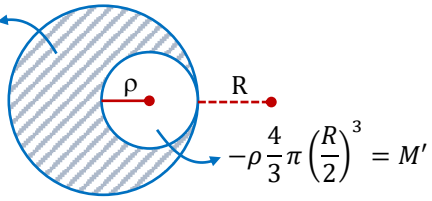
Illustration 18:

GFI at *P* = ?



Solution:

$\rho\frac{4}{3}\pi R^3=M$



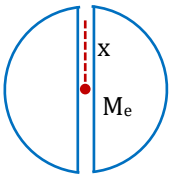
$-\rho\frac{4}{3}\pi\left(\frac{R}{2}\right)^3=M'$

$$E_P=\frac{GM}{(2R)^2}+\frac{GM'}{(3R/2)^2}$$

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Illustration 19:

A particle is released at a distance x from centre of Earth as shown in figure. Show that it performs SHM and find T.



Solution:

$$F = m \times GFI$$

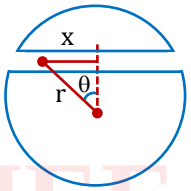
$$ma = m\left(G\frac{M_e x}{R^3}\right)$$

$$a = \left[\frac{GM_e}{R^3}\right]x \quad (\omega^2)$$

$$\boxed{T = 2\pi\sqrt{\frac{R^3}{GM_e}}}$$

Illustration 20:

A particle is released in a tunnel across Earth as shown in figure. Show that it performs SHM and find T.



Solution:

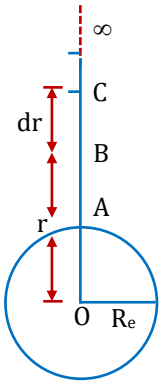
$$F_x = \left(\frac{GM_e r}{R^3}\sin\theta\right)m = ma$$

$$a = \underbrace{\left(\frac{GM_e}{R^3}\right)}_{\omega^2}x$$

$$\boxed{T = 2\pi\sqrt{\frac{R^3}{GM_e}}} \text{ (as } x = r\sin\theta)$$

Gravitational Potential Energy

Gravitational potential energy of two mass system is equal to the work done by an external agent in assembling them, while their initial separation was infinity. Consider a body of mass m placed at a distance r from another body of mass M. The gravitational force of attraction between them is given by, $F = \frac{GMm}{r^2}$



Now, Let the body of mass m is displaced from point. C to B through a distance $'dr'$ towards the mass M, then work done by internal conservative force (gravitational) is given by,

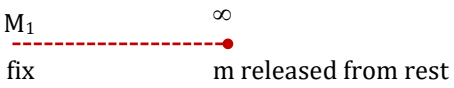
$$dW = F\,dr = \frac{GMm}{r^2}dr \quad \Rightarrow \quad \int dW = \int_{\infty}^{r}\frac{GMm}{r^2}dr$$

$\therefore$ Gravitational potential energy, $\boxed{U = -\frac{GMm}{r}}$

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Illustration 21:

Mass M_1 is fixed and mass, m is released from rest at infinity, find its velocity, when separation between them is 'r'.



Solution:

$$W_G + \underset{\Downarrow}{W_{ext}} = KE_f - \underset{\Downarrow}{KE_i}$$
$$0 \qquad\quad 0$$

$$U_i - U_f = KE_f - 0$$

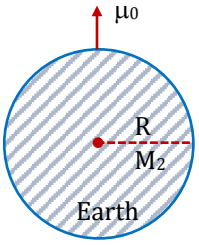
$$-\frac{GM_1m}{\infty} - \left(-\frac{GM_1m}{r}\right) = \frac{1}{2}mv^2 - 0$$

$$\frac{GM_1m}{r} = \frac{1}{2}mv^2$$

$$\boxed{v = \sqrt{\frac{2GM_1}{r}}}$$

Illustration 22:

Particle is projected vertically with speed μ_0, particle reaches maximum height of $3R$, find μ_0 (where R is radius of Earth)



Solution:

Height is always measured from surface.

$$W_G = \underset{\Downarrow}{KE_f} - KE_i$$
$$0$$

$$U_i - U_f = 0 - KE_i$$

$$-\frac{GM_em}{R} - \left(-\frac{GM_em}{4R}\right) = -\frac{1}{2}m\mu_0^2$$

$$-\frac{GM_em}{R} + \frac{GM_em}{4R} = -\frac{1}{2}m\mu_0^2$$

$$\frac{1}{2}m\mu_0^2 = \frac{3GM_em}{4R}$$

$$\mu_0 = \sqrt{\frac{3GM_e}{2R}}$$

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Illustration 23:

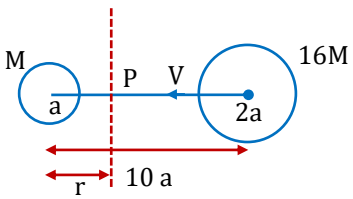
Distance between centres of two stars is 10 a. The masses of these stars are M and 16 M and their radii are a and $2a$ respectively. A body is fired straight from the surface of the larger star towards the smaller star. What should be its minimum initial speed to reach the surface of the smaller star?

Solution:

Let P be the point on the line joining the centres of the two planets and the net field at it is zero

Then, $\frac{GM}{r^2} - \frac{G.16M}{(10a-r)^2} = 0 \quad \Rightarrow \quad (10\,a-r)^2 = 16\,r^2$

$\Rightarrow \quad 10a - r = 4r \quad \Rightarrow \quad r = 2a$



Potential at point P,

$$V_P = \frac{-GM}{r} - \frac{G.16M}{(10a-r)^2} = \frac{-GM}{2a} - \frac{2GM}{a} = \frac{-5GM}{2a}$$

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Now if the particle projected from the larger planet has enough energy to cross this point, it will reach the smaller planet.

For this, the *K.E.* imparted to the body must be just enough to raise its total mechanical energy to a value which is equal to *P.E.* at point *P*.

i.e., $\frac{1}{2}mv^2 - \frac{G(16M)m}{2a} \frac{-GMm}{8a} = mV_P$

or $\frac{v^2}{2} - \frac{8GM}{a} - \frac{GM}{8a} = -\frac{5GMm}{2a}$

or $v^2 = \frac{45GM}{4a}$

or $v_{\min} = \frac{3}{2}\sqrt{\frac{5GM}{a}}$

Gravitational Self-Energy

The gravitational self-energy of a body (or a system of particles) is defined as the work done by an external agent in assembling the body (or system of particles) from infinitesimal elements (or particles) that are initially an infinite distance apart.

Gravitational Self Energy of a System of n Particles

Potential energy of *n* particles at an average distance *'r'* due to their mutual gravitational attraction is equal to the sum of the potential energy of all pairs of particle, i.e.,

$$U_s = -G \sum_{\substack{\text{all pairs} \\ j \neq i}} \frac{m_i m_j}{r_{ij}}$$

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This expression can be written as

$$U_s = -\frac{1}{2}G\sum_{i=1}^{i=n} \sum_{\substack{j=1 \\ j \neq i}}^{j=n} \frac{m_i m_j}{r_{ij}}$$

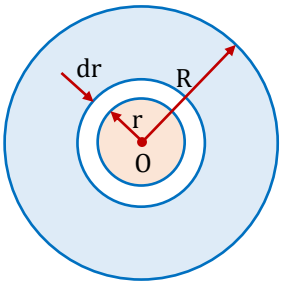
If consider a system of *'n'* particles, each of same mass *'m'* and separated from each other by the same average distance *'r'*, then self energy

or $U_s = -\frac{1}{2}G\sum_{i=1}^{n} \sum_{\substack{j=1 \\ j \neq i}}^{n} \left(\frac{m^2}{r}\right)_{ij}$

Thus, on the right hand side *'i'* comes *'n'* times while *'j'* comes $(n-1)$ times. Thus $U_s = -\frac{1}{2}Gn(n-1)\frac{m^2}{r}$.

Gravitational Self Energy of a Uniform Sphere (Star)

$$U_{sphere} = -G\frac{\left(\frac{4}{3}\pi r^3\rho\right)\left(4\pi r^2 dr\rho\right)}{r} \text{ where } \rho = \frac{M}{\left(\frac{4}{3}\right)\pi R^3} = -\frac{1}{3}G\left(4\pi\rho\right)^2 r^4 dr$$



$$U_{star} = -\frac{1}{3}G\left(4\pi\rho\right)^2 \int_0^R r^4 dr = -\frac{1}{3}G\left(4\pi\rho\right)^2 \left[\frac{r^5}{5}\right]_0^R = -\frac{3}{5}G\left(\frac{4\pi}{3}R^3\rho\right)^2 \frac{1}{R}$$

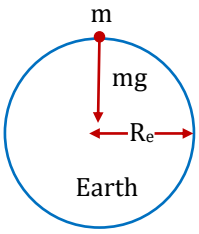
$\therefore$ $U_{star} = -\frac{3}{5}\frac{GM^2}{R}$

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Acceleration Due to Gravity

It is the acceleration, a freely falling body near the Earth's surface acquires due to the Earth's gravitational pull. The property by virtue of which a body experiences or exerts a gravitational pull on another body is called **gravitational mass** m_G, and the property by virtue of which a body opposes any change in its state of rest or uniform motion is called its **inertial mass** m_I, thus if $\vec{E}$ is the gravitational field intensity due to the Earth at *a* point *P*, and $\vec{g}$ is acceleration due to gravity at the same point, then $m_I\vec{g} = m_G\vec{E}$.

Now the value of inertial and gravitational mass happen to be exactly same to a great degree of accuracy for all bodies. Hence, $\vec{g} = \vec{E}$



The gravitational field intensity on the surface of Earth is therefore numerically equal to the acceleration due to gravity (g), there. Thus, we get,

where, M_e = Mass of Earth

R_e = Radius of Earth

Note:

Here the distribution of mass in the Earth is taken to be spherical and symmetrical so that its entire mass can be assumed to be concentrated at its center for the purpose of calculation of g.

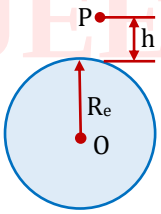
Variation of Acceleration Due to Gravity

(a) Effect of Altitude:

Acceleration due to gravity on the surface of the Earth is given by, $g = \frac{GM_e}{R_e^2}$.

Now, consider the body at a height *'h'* above the surface of the Earth, then the acceleration due to gravity at height *'h'* is given by

$$g_h = \frac{GM_e}{(R_e + h)^2} = g\left(1 + \frac{h}{R_e}\right)^{-2} \simeq g\left(1 - \frac{2h}{R_e}\right) \text{ when } h << R.$$

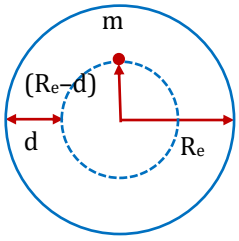


The decrease in the value of *'g'* with height $h = g - g_h = \frac{2gh}{R_e}$.

Then percentage decrease in the value of $'g' = \frac{g - g_h}{g} \times 100 = \frac{2h}{R_e} \times 100\%$.

(b) Effect of Depth:

The gravitational pull on the surface is equal to its weight i.e. $mg = \frac{GM_e m}{R_e^2}$



$$\therefore \quad mg = \frac{G \times \frac{4}{3}\pi R_e^3 \rho m}{R_e^2} \text{ or } g = \frac{4}{3}\pi G R_e \rho \qquad \text{...(i)}$$

When the body is taken to a depth *d*, the mass of the sphere of radius $(R_e - d)$ will only be effective for the gravitational pull and the outward shall will have no resultant effect on the mass. If the acceleration due to gravity on the surface of the solid sphere is g_d, then

$$g_d = \frac{4}{3}\pi G(R_e - d)\rho \qquad \text{...(ii)}$$

By dividing equation (ii) by equation (i)

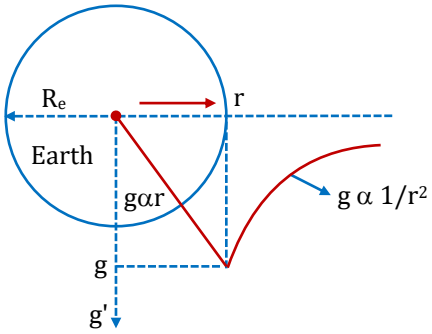
$$\Rightarrow \quad g_d = g\left(1 - \frac{d}{R_e}\right)$$

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Important Points:

(i) At the center of the Earth, $d = R_e$, so $g_{centre} = g\left(1-\frac{R_e}{R_e}\right)=0$.

Thus weight (mg) of the body at the centre of the Earth is zero.



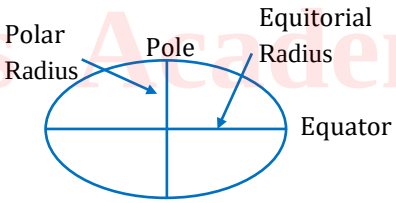
(ii) Percentage decrease in the value of 'g' with the depth = $\left(\frac{g-g_d}{g}\right)\times 100=\frac{d}{R_e}\times 100$.

(c) Effect of the Surface of Earth

The equatorial radius is about 21 *km* longer than its polar radius.

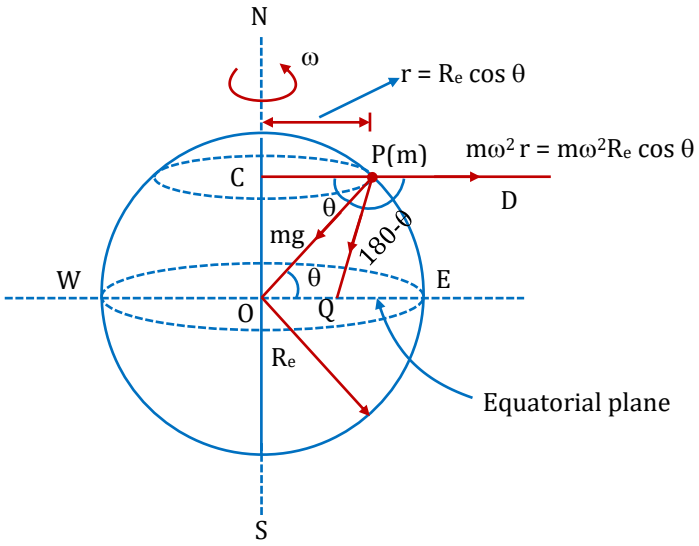
We know, $g=\frac{GM_e}{R_e^2}$. Hence $g_{pole} > g_{equator}$. The weight of the body increases as the body taken from the equator to the pole.

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(d) Effect of Rotation of the Earth

The Earth rotates around its axis with angular velocity ω. Consider a particle of mass m at latitude θ. The angular velocity of the particle is also ω.



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According to parallelogram law of vector addition, the resultant force acting on mass m along PQ is

$$F = [(mg)^2 + (m\omega^2 R_e \cos\theta)^2 + \{2mg \times m\omega^2 R_e \cos\theta\} \cos(180 - \theta)]^{1/2}$$

$$= [(mg)^2 + (m\omega^2 R_e \cos\theta)^2 - (2m^2 g\omega^2 R_e \cos\theta)\cos\theta]^{1/2}$$

$$= mg\left[1 + \left(\frac{R_e\omega^2}{g}\right)^2 \cos^2\theta - 2\frac{R_e\omega^2}{g}\cos^2\theta\right]$$

At pole $\theta = 90° \Rightarrow g_{pole} = g$, At equator $\theta = 0 \Rightarrow g_{equator} = g\left[1 - \frac{R_e\omega^2}{g}\right]$.

Hence $g_{pole} > g_{equator}$

If the body is taken from pole to the equator, then $g' = g\left[1 - \frac{R_e\omega^2}{g}\right]$.

Hence % change in weight = $\frac{mg - mg\left(1 - \frac{R_e\omega^2}{g}\right)}{mg} \times 100 = \frac{mR_e\omega^2}{mg} \times 100 = \frac{R_e\omega^2}{g} \times 100$

Key Points:

1. Apparent weight is equal to normal reaction is equal to reading of weighing machine is equal to $m \times g_{apparent}$.
   [Apparent weight = Normal reaction = Reading of weighing machine = $m \times g_{apparant}$]
2. Motion of an object is referred as free fall if its acceleration is same as acceleration due to gravity.

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Illustration 24:

At some planet, g_{app} at equator is equal to half of g_{app} at poles.

Find ω in terms of g and R.

Solution:

$$g_e = \frac{1}{2}g_p \quad \Rightarrow \quad g - R\omega^2 = \frac{1}{2}g \Rightarrow \omega^2 R = \frac{g}{2} \quad \Rightarrow \quad \omega = \sqrt{\frac{g}{2R}}$$

Weightlessness

State of the free fall $\left(\vec{a} = -\frac{GM}{r^2}\vec{r}\right)$ is called state of weightlessness. If a body is in a satellite (which does not produce its own gravity) orbiting the Earth at a height h above its surface then

True weight = $mg_h = \frac{mGM}{(R+h)^2} = \frac{mg}{\left(1 + \frac{h}{R}\right)^2}$; Apparent weight = $m(g_h - a)$

But, $a = \frac{v_0^2}{r} = \frac{GM}{r^2} = \frac{GM}{(R+h)^2} = g_h$; Apparent weight = $m(g_h - g_h) = 0$

Note:

The condition of weightlessness can be overcome by creating artificial gravity by rotating the satellite in addition to its revolution.

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Illustration 25:

Find ratio of gravitational field on the surface of two planets which are of uniform mass density and have radius R_1 and R_2 if

(a) they are of same mass (b) they are of same density.

Ans. (a) $\frac{g_1}{g_2}=\frac{R_2^2}{R_1^2}$ (b) $\frac{g_1}{g_2}=\frac{R_1}{R_2}$

Solution:

$$g=\frac{GM}{R^2}=\frac{G\frac{4}{3}\pi R^3\rho}{R^2}=\left(\frac{4\pi GR}{3}\right)\rho$$

Escape Speed

The minimum speed required to send a body out of the gravity field of a planet (send it to $r \to \infty$).

Escape Speed at Earth's Surface

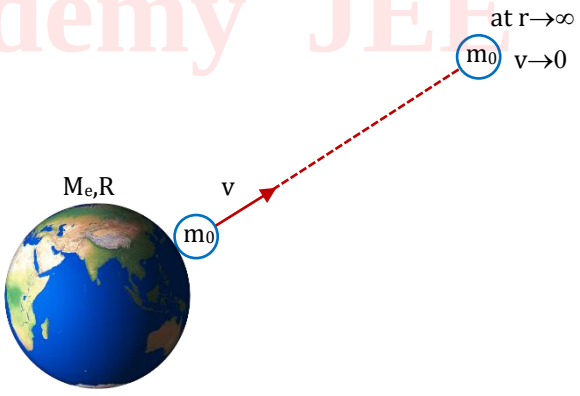
Suppose a particle of mass m is on Earth's surface. We project it with a velocity V from the Earth's surface, so that it just reaches $r \to \infty$ (at $r \to \infty$, its velocity become zero). Applying energy conservation between initial position (when the particle was at Earth's surface) and find positions (when the particle just reaches to $r \to \infty$)

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$$K_i + U_i = K_f + U_f$$

$$\frac{1}{2}m_0v^2+m_0\left(-\frac{GM_e}{R}\right)=0+m_0\left(-\frac{GM_e}{(r\to\infty)}\right)$$

$$\Rightarrow \quad v=\sqrt{\frac{2GM_e}{R}}$$



Escape speed from Earth is surface $v_e=\sqrt{\frac{2GM_e}{R}}$

If we put the values of G, Me, R then we get $V_e = 11.2\ km/s$.

Escape Speed Depends on:

(i) Mass (M_e) and size (R) of the planet.

(ii) Position from where the particle is projected.

Escape speed does not depend on:

(i) Mass of the body which is projected (m_0)

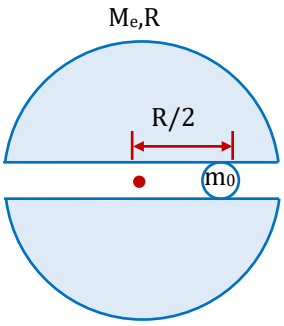
(ii) Angle of projection.

If a body is thrown from Earth's surface with escape speed, it goes out of Earth's gravitational field and never returns to the Earth's surface. But it starts revolving around the sun.

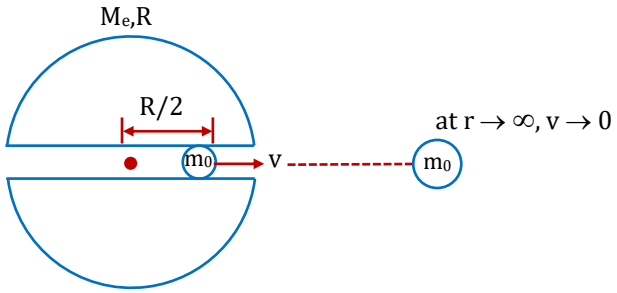
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Illustration 26:

A very small groove is made in the Earth, and a particle of mass m_0 is placed at $R/2$ distance from the centre. Find the escape speed of the particle from that place.



Solution:



Suppose we project the particle with speed v, so that it just reaches at $(r \to \infty)$.

Applying energy conservation $K_i + U_i = K_f + U_f$

$$\frac{1}{2}m_0v^2 + m_0\left(-\frac{GM_e}{2R^3}\left(3R^2 - \left(\frac{R}{2}\right)^2\right)\right) = 0 + 0$$

$$v = \sqrt{\frac{11GM_e}{4R}} = v_e \text{ at that position.}$$

Illustration 27:

Find radius of such planet on which the man escapes through jumping. The capacity of jumping of person on Earth is 1.5 m. Density of planet is same as that of Earth.

Solution:

For a planet: $\frac{1}{2}mv^2 - \frac{GM_Pm}{R_p} = 0 \qquad \Rightarrow \frac{1}{2}mv^2 = \frac{GM_Pm}{R_p}$

On Earth $\to \frac{1}{2}mv^2 = m\left(\frac{GM_E}{R_E^2}\right)h$

$\therefore \qquad \frac{GM_pm}{R_p} = \frac{GM_E.m}{R_E^2}\cdot h \Rightarrow \frac{M_p}{R_p} = \frac{M_Eh}{R_E^2}$

$\therefore$ Density (ρ) is same $\Rightarrow \frac{4/3\pi R_p^3\rho}{R_p} = \frac{4/3\pi R_E^3\rho}{R_E^2}h \Rightarrow R_P = \sqrt{R_Eh}$

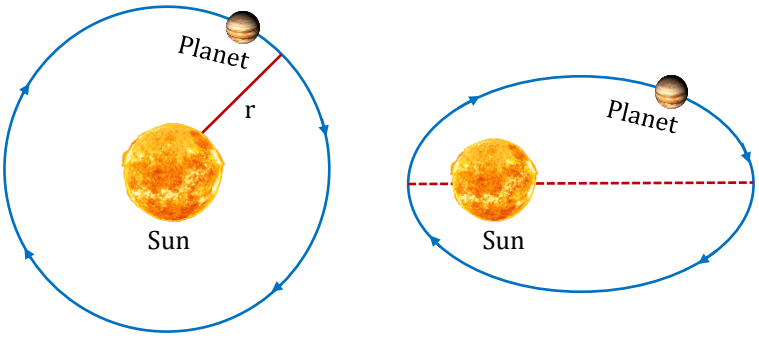
Kepler's Law for Planetary Motion

Suppose a planet is revolving around the sun, or a satellite is revolving around the Earth, then the planetary motion can be studied with help of Kepler's three laws.

Kepler's Law of Orbit

Each planet moves around the Sun in a circular path or elliptical path with the Sun at its focus. (In fact circular path is a subset of elliptical path)

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Law of Areal Velocity

To understand this law, lets understand the angular momentum conservation for the planet.

If a planet moves in an elliptical orbit, the gravitation force acting on it always passes through the centre of the sun. So, torque of this gravitation force about the centre of the Sun will be zero. Hence, we can say that angular momentum of the planet about the centre of the Sun will remain conserved (constant)

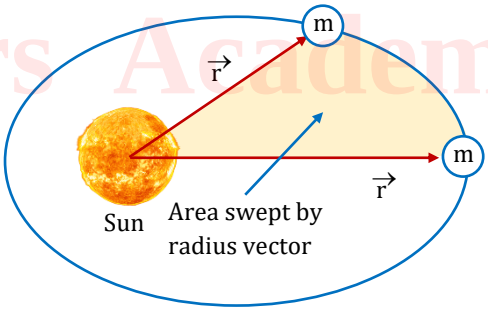
τ about the Sun = 0

$\Rightarrow \quad \frac{dL}{dt} = 0 \quad \Rightarrow \quad L_{\text{planet}}$ / Sun = constant $\quad \Rightarrow \quad mvr \sin\theta$ = constant

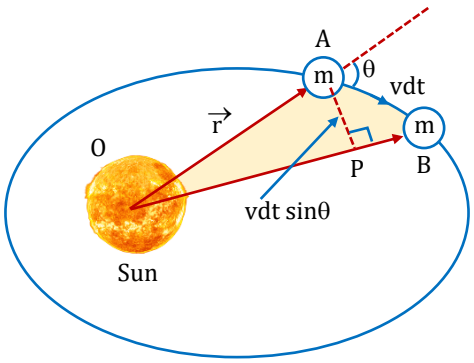
Now we can easily study the Kepler's law of aerial velocity.

If a planet moves around the sun, the radius vector ($\vec{r}$) also rotates and sweeps area as shown in figure. Now let's find rate of area swept by the radius vector ($\vec{r}$).

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Suppose a planet is revolving around the Sun and at any instant its velocity is v, and angle between radius vector ($\vec{r}$) and velocity ($\vec{v}$). In dt time, it moves by a distance vdt, during this dt time, area swept by the radius vector will be OAB which can be assumed to be a triangle



dA = 1/2 (Base) (Perpendicular height)

dA = 1/2 (r) (vdt sin θ)

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So, rate of area swept $\frac{dA}{dt}=\frac{1}{2}vr\sin\theta$

We can write $\frac{dA}{dt}=\frac{1}{2}\frac{mvr\sin\theta}{m}$

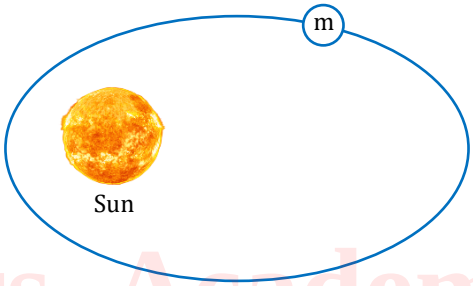
Where $mvr\sin\theta$ = Angular momentum of the planet about the sun, which remains conserved (constant)

$\Rightarrow \quad \frac{dA}{dt}=\frac{L_{planet/sun}}{2m}=\text{constant}$

So, **rate of area swept by the radius vector is constant.**

Illustration 28:

Suppose a planet is revolving around the Sun in an elliptical path given by $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Find time period of revolution. Angular momentum of the planet about the Sun is L.



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Solution:

Rate of area swept $\frac{dA}{dt}=\frac{L}{2m}$ = constant

$\Rightarrow \quad dA=\frac{L}{2m}dt\ ; \int_{A=0}^{A=\pi ab} dA=\int_{t=0}^{t=T}\frac{L}{2m}dt \quad \Rightarrow \quad \pi ab=\frac{L}{2m}T \quad \Rightarrow T=\frac{2\pi mab}{L}$

Kepler's Law of Time Period

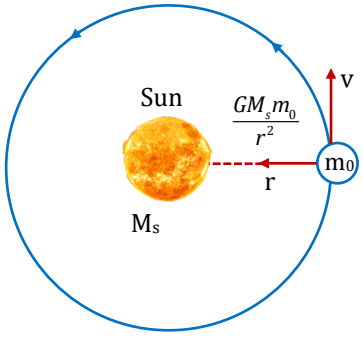
Suppose a planet is revolving around the Sun in circular orbit then $\frac{m_0v^2}{r}=\frac{GM_Sm_0}{r^2}$

$$v=\sqrt{\frac{GM_s}{r}}$$

Time period of revolution is $T=\frac{2\pi r}{v}=2\pi r\sqrt{\frac{r}{GM_s}}$

$$T^2=\left(\frac{4\pi^2}{GM_s}\right)r^3 \quad \Rightarrow \ T^2=\alpha r^3$$

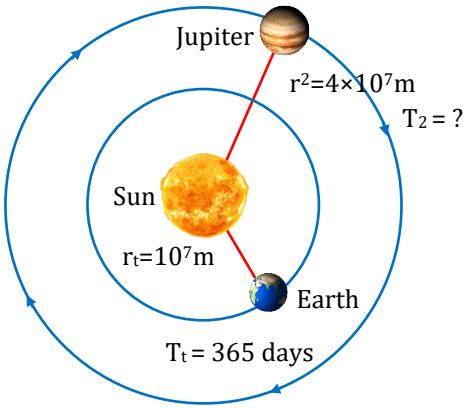
For all the planet of a Sun, $T^2\propto r^3$



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Illustration 29:

The Earth and Jupiter are two planets of the sun. The orbital radius of the Earth is 10^7 *m* and that of Jupiter is 4×10^7 *m*. If the time period of revolution of Earth is $T = 365$ days, find time period of revolution of the Jupiter.



Solution:

For both the planets

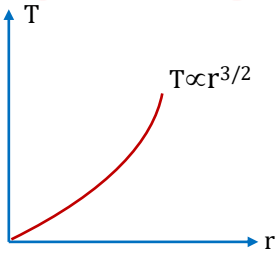
$$T^2 \propto r^3$$

$$\left(\frac{T_{\text{Jupiter}}}{T_{\text{Earth}}}\right)^2 = \left(\frac{r_{\text{Jupiter}}}{r_{\text{Earth}}}\right)^3$$

$$\Rightarrow \quad \left(\frac{T_{\text{Jupiter}}}{365\,\text{days}}\right)^2 = \left(\frac{4\times10^7}{10^7}\right)^3$$

$T_{\text{jupiter}} = 8 \times 365$ days

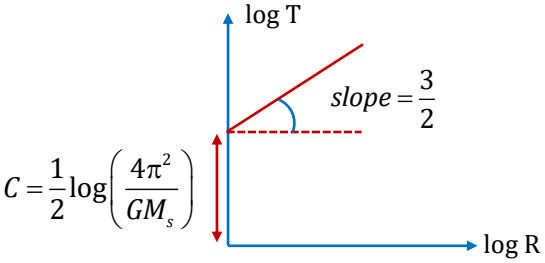
Graph of T vs r :



Graph of log T v/s log R :

$$T^2 = \left(\frac{4\pi^2}{GM_s}\right)R^3 \Rightarrow 2log\ T = log\left(\frac{4\pi^2}{GM_s}\right) + 3log\ R$$

$$\log T = \frac{1}{2}\log\left(\frac{4\pi^2}{GM_s}\right) + \frac{3}{2}\log R$$



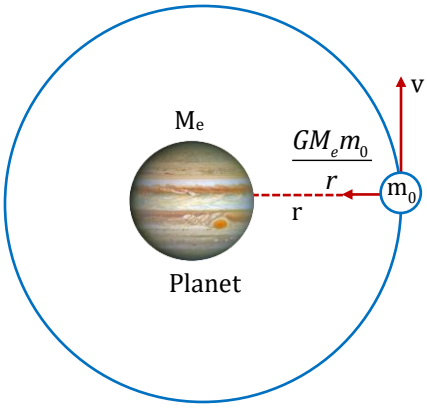
- If planets are moving in elliptical orbit, then $T^2 \propto a^3$ where a = semi major axis of the elliptical path.

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Circular Motion of a Satellite Around a Planet

Suppose at satellite of mass m_0 is at a distance r from a planet. If the satellite does not revolve, then due to the gravitational attraction, it may collide to the planet.

To avoid the collision, the satellite revolve around the planet, for circular motion of satellite.



$$\Rightarrow \quad \frac{GM_em_0}{r^2}=\frac{m_0v^2}{r} \qquad \text{....(i)}$$

$$\Rightarrow \quad v=\sqrt{\frac{GM_e}{r}}$$ this velocity is called orbital velocity (v_0)

$$v_0=\sqrt{\frac{GM_e}{r}}$$

Total Energy of the Satellite Moving in Circular Orbit

(i) $KE=\frac{1}{2}m_0v^2$ and from equation (i)

$$\frac{m_0v^2}{r}=\frac{GM_em_0}{r^2}$$

$$\Rightarrow \quad m_0v^2=\frac{GM_em_0}{r}$$

$$\Rightarrow \quad KE=\frac{1}{2}m_0v^2=\frac{GM_em_0}{2r}$$

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(ii) Potential energy $U=-\frac{GM_em_0}{r}$

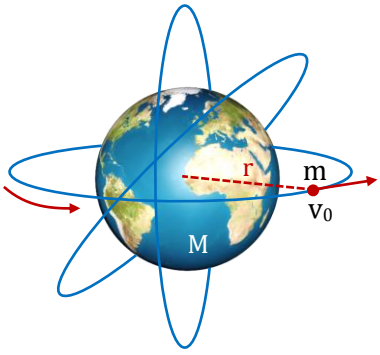
Total energy = $KE+PE=\left(\frac{GM_em_0}{2r}\right)+\left(\frac{-GM_em_0}{r}\right)$

$$TE=-\frac{GM_em_0}{2r}$$

Total energy is negative. It shows that the satellite is still bounded with the planet.

Essential Condition's for Satellite Motion

- Centre of satellite's orbit coincide with centre of Earth.
- Plane of orbit of satellite is passing through centre of Earth.



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Geo-Stationary Satellite

We know that the Earth rotates about its axis with angular velocity ω_{Earth} and time period T_{Earth} = 24 hours.

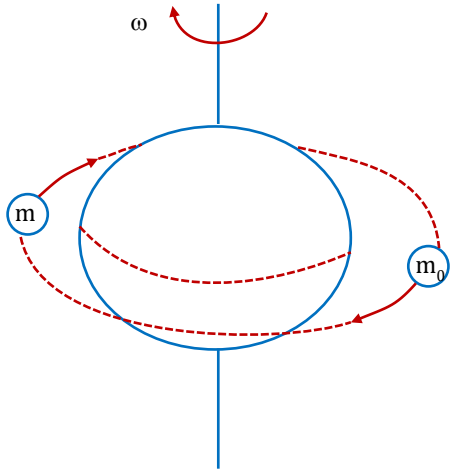
Suppose a satellite is set in an orbit which is in the plane of the equator, whose ω is equal to ω_{Earth}, (or its T is equal to T_{Earth} = 24 hours) and direction is also same as that of Earth. Then as seen from Earth, it will appear to be stationary. This type of satellite is called geo- stationary satellite. For a geo-stationary satellite,

$w_{satellite} = w_{Earth}$

$\Rightarrow$ $T_{satellite} = T_{Earth}$ = 24 hr.

So, time period of a geo-stationery satellite must be 24 hours. To achieve T = 24 hour, the orbital radius geo-stationary satellite:

$$T^2 = \left(\frac{4\pi^2}{GM_e}\right) r^3$$

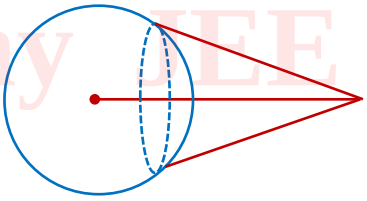


Putting the values, we get orbital radius of geo stationary satellite r = 6.6 R_e (here R_e = radius of the Earth)

height from the surface h = 5.6 R_e.

Special Points about Geo–Stationary Satellite

- All three essential conditions for satellite motion should be followed.
- It rotates in equatorial plane.
- Its height from Earth surface is 36000 *km*. ($\sim 6R_e$)
- Its angular velocity and time period should be same as that of Earth.
- Its rotating direction should be same as that of Earth (West to East).



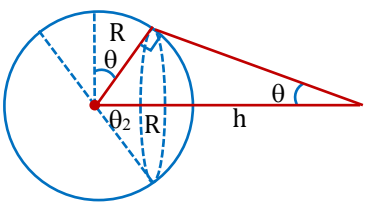
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- Its orbit is called parking orbit and its orbital velocity is 3.1 *km*./sec.
- Maximum latitude at which message can be received by geostationary satellite is $\theta = \cos^{-1}\left(\frac{R_e}{R_e + h}\right)$
- The area of Earth's surface covered by geostationary satellite is $S = \Omega R_e^2 = \frac{2\pi h R_e^2}{R_e + h}$
- The area of Earth's surface escaped by geostationary satellite is
- The area of Earth's surface escaped by geostationary satellite is

$$\cos\theta_2 = \frac{R}{R+h}$$

$$4\pi R^2 - 2\pi R^2(1-\cos\theta_2) = 2\pi r^2 + 2\pi r^2\cos(90^\circ - \theta)$$

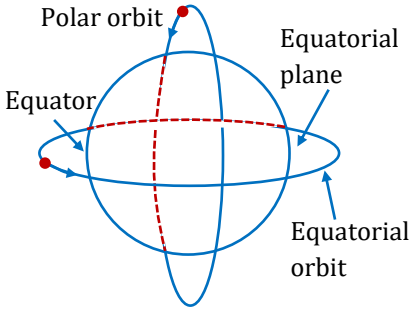
$$= 2\pi R^2(1+\sin\theta)$$



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Polar Satellite (Sun–Synchronous Satellite)

It is that satellite which revolves in polar orbit around Earth. A polar orbit is that orbit whose angle of inclination with equatorial plane of Earth is 90° and a satellite in polar orbit will pass over both the north and south geographic poles once per orbit. Polar satellites are Sun–synchronous satellites. Every location on Earth lies within the observation of polar satellite twice each day. The polar satellites are used for getting the cloud images, atmospheric data, ozone layer in the atmosphere and to detect the ozone hole over Antarctica.



Only the equatorial orbits are stable for a satellite. For any satellite to orbit around in a stable orbit, it must move in such an orbit so that the centre of Earth lies at the centre of the orbit.

Binding Energy

Total mechanical energy (potential + kinetic) of a closed system is negative. The modulus of this total mechanical energy is known as the binding energy of the system. This is the energy due to which system is closed or different parts of the system are bound to each other.

Binding Energy of Satellite (System):

$$B.E. = -T.E.$$

$$B.E. = \frac{1}{2}mv_0^2 = \frac{GMm}{2r} = \frac{L^2}{2mr^2}$$

Hence $B.E. = K.E. = -T.E. = \frac{-\text{P.E.}}{2}$

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Work Done in Changing the Orbit of Satellite:

W = Change in mechanical energy of system but $E = \frac{-GMm}{2r}$, so $W = E_2 - E_1 = \frac{GMm}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$

Illustration 30:

A satellite moves eastwards very near to the surface of the Earth in equatorial plane with speed (v_0). Another satellite moves at the same height with the same speed in the equatorial plane but westwards. If R = radius of the Earth and ω be its angular speed of the Earth about its own axis. Then find the approximate difference in the two time period as observed on the Earth.

Solution:

$$T_{west} = \frac{2\pi R}{v_0 + R\omega}$$

and $$T_{east} = \frac{2\pi R}{v_0 - R\omega}$$

$$\Rightarrow \quad \Delta T = T_{east} - T_{west} = 2\pi R\left[\frac{2R\omega}{v_0^2 - R^2\omega^2}\right] = \frac{4\pi\omega R^2}{v_0^2 - R^2\omega^2}$$

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Illustration 31:

Find W_{ext} to change orbital radius of satellite from r_1 to r_2.

Solution:

$$W_{ext} + W_G = KE_f - KE_i$$

$$W_{ext} + U_i - U_f = KE_f - KE_i$$

$$W_{ext} = (K_f + U_f) - (K_i + U_i) = TE_f - TE_i$$

$$W_{ext} = \frac{-GM_pM_S}{2r_2} - \left(\frac{-GM_pM_S}{2r_1}\right)$$

Illustration 32:

An artificial satellite (mass m) of a planet (mass M) revolves in a circular orbit whose radius is n times the radius R of the planet. In the process of motion, the satellite experiences a slight resistance due to cosmic dust. Assuming the force of resistance on satellite to depend on velocity as $F = av^2$ where $'a'$ is a constant, calculate how long the satellite will stay in the space before it falls onto the planet's surface.

Solution:

Air resistance $F = -av^2$, where orbital velocity $v = \sqrt{\frac{GM}{r}}$

r = The distance of the satellite from planet's centre $\Rightarrow F = -\frac{GMa}{r}$

The work done by the resistance force $dW = Fdx = Fvdt = \frac{GMa}{r}\sqrt{\frac{GM}{r}}dt = \frac{(GM)^{3/2}a}{r^{3/2}}dt$ ...(i)

The loss of energy of the satellite = dE $\therefore \frac{dE}{dr} = \frac{d}{dr}\left[-\frac{GM\ m}{2r}\right] = \frac{GMm}{2r^2}$ $\Rightarrow dE = \frac{GMm}{2r^2}dr$ ...(ii)

Since, $dE = -dW$ (work energy theorem) $-\frac{GMm}{2r^2}dr = \frac{(GM)^{3/2}}{r^{3/2}}adt$

$$\Rightarrow \quad t = -\frac{m}{2a\sqrt{GM}}\int_{nR}^{R}\frac{dr}{\sqrt{r}} = \frac{m\sqrt{R}\left(\sqrt{n}-1\right)}{a\sqrt{GM}} = \left(\sqrt{n}-1\right)\frac{m}{a\sqrt{gR}}$$

Illustration 33:

Two satellites S_1 and S_2 revolve round a planet in coplanar circular orbits in the same sense. Their periods of revolution are 1 h and 8h respectively. The radius of the orbit of S_1 is 10^4 km. When S_2 is closest to S_1, find:

(a) the speed of S_2 relative to S_1 and

(b) the angular speed of S_2 as observed by an astronaut in S_1.

Solution:

Let the mass of the planet be M, that of S_1 be m_1 and S_2 be m_2. Let the radius of the orbit of S_1 be R_1(= 10^4 km) and of S_2 be R_2.

Let v_1 and v_2 be the linear speeds of S_1 and S_2 with respect to the planet. Figure shows the situation.

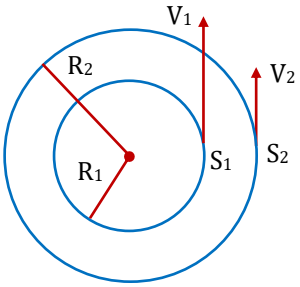
As the square of the time period is proportional to the cube of the radius.

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$$\left(\frac{R_2}{R_1}\right)^3 = \left(\frac{T_2}{T_1}\right)^2 = \left(\frac{8h}{1h}\right)^2 = 64$$

or, $\frac{R_2}{R_1} = 4$

or, $R_2 = 4R_1 = 4 \times 10^4\ m.$



Now the time period of S_1 is 1 h. So,

$$\frac{2\pi R_1}{v_1} = 1h$$

or, $v_1 = \frac{2\pi R_i}{1h} = 2\pi \times 10^4\ km/h$

Similarly, $v_2 = \frac{2\pi R_2}{8h} = \pi \times 10^4\ km/h$

(a) At the closest separation, they are moving in the same direction. Hence the speed of S_2 with respect to S_1 is $|v_2 - v_1| = \pi \times 10^4\ km/h.$

(b) As seen from S_1, the satellite S_2 is at a distance $R_2 - R_1 = 3 \times 10^4\ km$ at the closest separation. Also, is moving at $\pi \times 10^4\ km/h$ in a direction perpendicular to the line joining them. Thus, the angular speed of S_2 as observed by S_1 is

$$\omega = \frac{\pi \times 10^4\ km/h}{3 \times 10^4\ km/h} = \frac{\pi}{3} rad/h.$$

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Illustration 34:

A space–ship is launched into a circular orbit close to the Earth's surface. What additional speed should now be imparted to the spaceship so that orbit to overcome the gravitational pull of the Earth.

Solution:

Let ΔK be the additional kinetic energy imparted to the spaceship to overcome the gravitation pull then by energy conservation

$$-\frac{GMm}{2R} + \Delta K = 0 + 0$$

$\Rightarrow$ $\Delta K = \frac{GMm}{2R}$

Total kinetic energy = $\frac{GMm}{2R} + \Delta K = \frac{GMm}{2R} + \frac{GMm}{2R} = \frac{GMm}{R}$

Then, $\frac{1}{2}mv_2^2 = \frac{GMm}{R}$

$\Rightarrow$ $v_2 = \sqrt{\frac{2GM}{R}}$

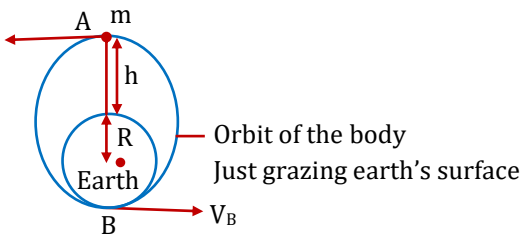
But, $v_1 = \sqrt{\frac{GM}{R}}$

So, Additional velocity = $v_2 - v_1 = \sqrt{\frac{2GM}{R}} - \sqrt{\frac{GM}{R}} = \left(\sqrt{2}-1\right)\sqrt{\frac{GM}{R}}$

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Illustration 35:

For a particle projected in a transverse direction from a height h above Earth's surface, find the minimum initial velocity so that it just grazes the surface of Earth path of this particle would be an ellipse with center of Earth as the farther focus, point of projection as the apogee and a diametrically opposite point on Earth's surface as perigee.



Solution:

Suppose velocity of projection at point A is v_A and at point B, the velocity of the particle is v_B.

then applying Newton's 2nd law at point A and B, we get, $\frac{mv_A^2}{r_A}=\frac{GM_em}{(R+h)^2}$ and $\frac{mv_B^2}{r_B}=\frac{GM_em}{R^2}$

Where r_A and r_B are radius of curvature of the orbit at points A and B of the ellipse,

But $r_A = r_B = r$(say).

Now applying conservation of energy at points A and B

$$\frac{-GM_em}{R+h}+\frac{1}{2}mv_A^2=\frac{-GM_em}{R}+\frac{1}{2}mv_B^2$$

$$\Rightarrow \quad GM_em\left(\frac{1}{R}-\frac{1}{(R+h)}\right)=\frac{1}{2}\left(mv_B^2-mv_A^2\right)=\left(\frac{1}{2}rGM_em\left(\frac{1}{R^2}-\frac{1}{(R+h)^2}\right)\right)$$

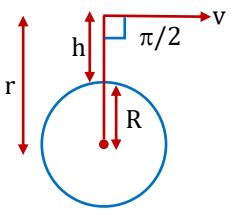
or, $$r=\frac{2R(R+h)}{2R+h}=\frac{2Rr}{R+r}$$

$$\therefore \quad V_A^2=\frac{rGM_e}{(R+h)^2}=2GM_e\frac{R}{r(r+R)}$$

where r = Distance of point of projection from Earth's centre = $R + h$.

Conditions for Different Trajectory

For a body being projected tangentially from above Earth's surface, say at a distance r from Earth's center, the trajectory would depend on the velocity of projection v.



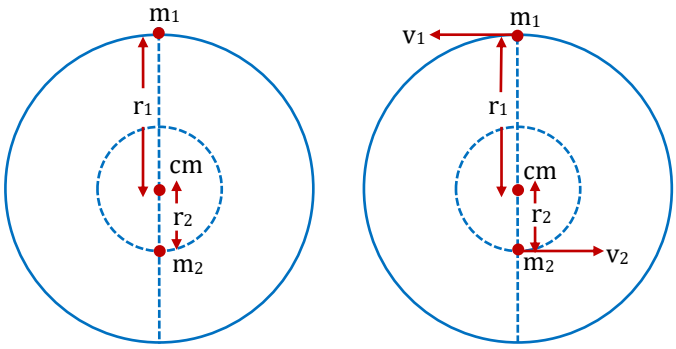
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	Velocity	Orbit
1.	Velocity, $v < \sqrt{\frac{GM}{r}\left(\frac{2R}{r+R}\right)}$	Body returns to Earth
2.	$\sqrt{\frac{GM}{r}} > v > \sqrt{\frac{GM}{r}\left(\frac{2R}{r+R}\right)}$	Body acquires an elliptical orbit with Earth as the far-focus w.r.t. the point of projection.
3.	Velocity is equal to the critical velocity of the orbit, $v = \sqrt{\frac{GM_e}{r}}$	Circular orbit with radius r
4.	Velocity is between the critical and escape velocity of the orbit $\sqrt{\frac{2GM_e}{r}} > v > \sqrt{\frac{GM_e}{r}}$ i.e $TE < 0$	Body acquires an elliptical orbit with Earth as the near focus w.r.t. the point of projection.
5.	$v = v_{esc} = \sqrt{\frac{2GM_e}{r}}$ i.e $TE = 0$	Body just escapes Earth's gravity, along a parabolic path.
6.	$v > v_{esc} = \sqrt{\frac{2GM_e}{r}}$ i.e $TE > 0$	Body escape Earth's gravity along a hyperbolic path.

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Double (Binary) Star System

If two stars of comparable mass revolve about their common centre of mass under the influence of mutual gravitational force then this system is called binary star system.



For centre of mass to remain at rest both the stars should rotate with same 'ω'.

Let distance of centre of mass for m_1 is r_1 and m_2 is r_2. Then

$m_1r_1 = m_2r_2$ ...(i)

$r_1 + r_2 = r$ ...(ii)

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From (i) and (ii)

$$r_1 = \frac{m_2 r}{m_1 + m_2}$$

$$r_2 = \frac{m_1 r}{m_1 + m_2}$$

$T_1 = T_2 \quad \Rightarrow \quad \omega_1 = \omega_2 \qquad ...(iii)$

$$\frac{Gm_1m_2}{r^2} = F = m_1\omega_1^2 r_1 \qquad ...(iv)$$

$$\frac{Gm_1m_2}{r^2} = m_2\omega_2^2 r_2 \qquad ...(v)$$

$\therefore$ $$\frac{Gm_1(m_1+m_2)}{r^2 m_1 r} = \omega_2^2$$

$$\frac{G(m_1+m_2)}{r^3} = \omega_1^2$$

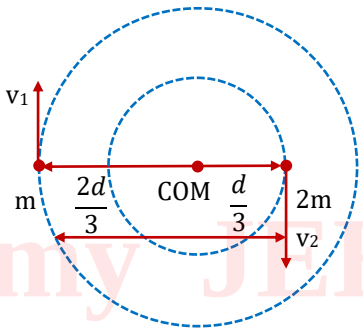
Angular momentum of system

$$L = m_1v_1r_1 + m_2v_2r_2$$

$$L = \left(\frac{m_1m_2r}{m_1+m_2}\right)\omega(r)$$

$$l = \mu\omega r^2$$

where $\mu = \frac{m_1m_2}{m_1+m_2}$ is the reduced mass of system.



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A pair of stars rotating about their COM (centre of mass)

$$F = \frac{GM(2m)}{d^2}$$

	Star-1	Star-2	Ratio
1.	$F = \frac{mv_1^2}{2d/3}$	$F = \frac{mv_1^2}{d/3}$	$\frac{v_1}{v_2} = \frac{2}{1}$
	$G\frac{M_2m}{d^2} = \frac{mv_1^2}{2d/3}$	$G\frac{M_2m}{d^2} = \frac{mv_2^2}{d/3}$	
	$v_1 = 2\sqrt{\frac{Gm}{3d}}$	$v_2 = \sqrt{\frac{Gm}{3d}}$	
2.	$T_1 = \frac{2\pi}{v_1}\frac{2d}{3}$	$T_2 = \frac{2\pi}{v_2}\frac{d}{3}$	$\frac{T_1}{T_2} = \frac{1}{1}$
3.	$KE_1 = \frac{1}{2}mv_1^2$	$KE_2 = \frac{1}{2}mv_2^2$	$\frac{KE_1}{KE_2} = \frac{2}{1}$
4.	$L_1 = mv_1\frac{2d}{3}$	$L_2 = 2mv_2\frac{d}{3}$	$\frac{L_1}{L_2} = \frac{2}{1}$

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Illustration 36:

In a double star, two stars (one of mass m and the other of $2m$) distant d apart rotate about their common centre of mass. Deduce an expression of the period of revolution. Show that the ratio of their angular momentum about the centre of mass is the same as the ratio of their kinetic energies.

Solution:

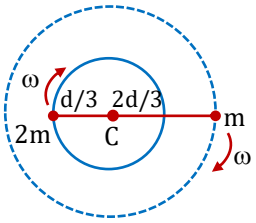
The centre of mass C will be at distances $d/3$ and $2d/3$ from the masses $2m$ and m respectively. Both the stars rotate round C in their respective orbits with the same angular velocity ω. The gravitational force acting on each star due to the other supplies the necessary centripetal force.

The gravitational force on either star is $\frac{G(2m)m}{d^2}$. If we consider the rotation of the smaller star, the centripetal force ($m r \omega^2$) is $\left[m\left(\frac{2d}{3}\right)\omega^2\right]$ and for bigger star $\left[\frac{2md\omega^2}{3}\right]$ i.e. same

$$\therefore \quad \frac{G(2m)m}{d^2} = m\left(\frac{2d}{3}\right)\omega^2 \quad \text{or} \quad \omega = \sqrt{\left(\frac{3Gm}{d^3}\right)}$$

Therefore, the period of revolution is given by

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\left(\frac{d^3}{3Gm}\right)}$$



The ratio of the angular momentum is

$$\frac{(I\omega)_{big}}{(I\omega)_{small}} = \frac{I_{big}}{I_{small}} = \frac{(2m)\left(\frac{d}{3}\right)^2}{m\left(\frac{2d}{3}\right)^2} = \frac{1}{2}$$

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Since ω is same for both. The ratio of their kinetic energies is $\frac{\left(\frac{1}{2}I\omega^2\right)_{big}}{\left(\frac{1}{2}I\omega^2\right)_{small}} = \frac{I_{big}}{I_{small}} = \frac{1}{2}$, which is the same as the ratio of their angular momentum.

Gravitational Pressure

Illustration 37:

A uniform sphere has a mass M and radius R. Find the pressure p inside the sphere, caused by gravitational compression, as a function of the distance r from its centre. Evaluate p at the centre of the Earth, assuming it to be a uniform sphere.

Ans. $p = \frac{3}{8}(1 - r^2/R^2)\, GM^2/\pi R^4$. About 1.8×10^8 atmospheres.

Solution:

Consider an element of a cylinder extending from centre to surface.

$$p = \int \frac{dmg}{A} = -\int_R^r \frac{GMr}{R^3} \times 4\pi r^2 \rho dr$$

$$= \frac{4\pi G\rho^2}{3}\frac{\left[R^2 - r^2\right]}{2} = \frac{2\pi G\rho^2}{3}\left[R^2 - r^2\right]$$

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Illustration 38:

A spherical body of radius R consists of a fluid of constant density and is in equilibrium under its own gravity. If $P(r)$ is the pressure at $r(r < R)$, then the correct option(s) is(are) :-

(A) $P(r = 0) = 0$

(B) $\frac{P(r=3R/4)}{P(r=2R/3)}=\frac{63}{80}$

(C) $\frac{P(r=3R/5)}{P(r=2R/5)}=\frac{16}{21}$

(D) $\frac{P(r=R/2)}{P(r=R/3)}=\frac{20}{27}$

Ans. (BC)

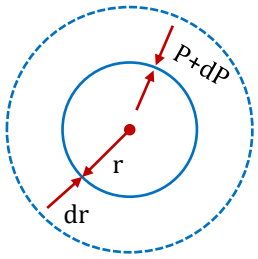
Solution:

$$dF = [P - (P + dP)]A$$

$$\Rightarrow \quad \frac{Gm}{r^2}dm=-(dP)A$$

$$\Rightarrow \quad \int_0^P dP=-\int_R^r \frac{G\frac{Mr^3}{R^3}4ar^2dr\rho}{r^2(4\pi r^2)}$$

$$\therefore \quad P=\frac{G\rho M}{2R}\left(1-\frac{r^2}{R^2}\right)$$



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